This cell initializes the TensorFlow environment, verifies GPU availability (Tesla T4), and enables dynamic GPU memory allocation to prevent memory overflow during training.

```
import os
import numpy as np
import pandas as pd
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint, ReduceLR
from tensorflow.keras.metrics import AUC, Precision, Recall
print("TensorFlow:", tf.__version__)
print("GPUs:", tf.config.list_physical_devices("GPU"))

import random

seed = 42
tf.random.set_seed(seed)
np.random.seed(seed)
random.seed(seed)
```

```
TensorFlow: 2.19.0
GPUs: [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]
```

```
!nvidia-smi
```

```
Sun Mar  1 19:13:31 2026
+-----------------------------------------------------------------------------------
| NVIDIA-SMI 580.82.07              Driver Version: 580.82.07      CUDA Version:
+-----------------------------------------+------------------------+-------------
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Un
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  C
|                                         |                        |
|=========================================+========================+=============
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |
| N/A   54C    P8             11W /   70W |       3MiB /  15360MiB |      0%
|                                         |                        |
+-----------------------------------------+------------------------+-------------

+-----------------------------------------------------------------------------------
| Processes:
|  GPU   GI   CI              PID   Type   Process name                        G
|        ID   ID                                                               U
|===================================================================================
|  No running processes found
+-----------------------------------------------------------------------------------
```

```
gpus = tf.config.list_physical_devices('GPU')
if gpus:
```

```
    try:
        for gpu in gpus:
            tf.config.experimental.set_memory_growth(gpu, True)
        print("GPU memory growth enabled")
    except RuntimeError as e:
        print(e)
```

```
GPU memory growth enabled
```

The HAM10000 dataset was stored in Google Drive and extracted into the Colab runtime environment. File integrity was verified by sampling image paths and checking existence.

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
!apt-get update -qq && apt-get install -y -qq unrar
```

```
W: Skipping acquire of configured file 'main/source/Sources' as repository 'http
```

```
!mkdir -p /content/HAM10000
!unrar x "/content/drive/MyDrive/HAM10000.rar" "/content/HAM10000/"
```

```
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034291.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034292.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034293.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034294.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034295.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034296.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034297.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034298.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034299.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034300.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034301.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034302.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034303.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034304.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034305.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034306.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034307.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034308.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034309.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034310.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034311.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034312.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034313.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034314.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034315.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034316.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034317.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034318.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034319.
Extracting  /content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034320.
Extracting  /content/HAM10000/archive (2)/HAM10000_metadata.csv       OK
All OK
```

```
!ls "/content/HAM10000/archive (2)/HAM10000_images_part_1" | head
```

```
ISIC_0024306.jpg
ISIC_0024307.jpg
ISIC_0024308.jpg
ISIC_0024309.jpg
ISIC_0024310.jpg
ISIC_0024311.jpg
ISIC_0024312.jpg
ISIC_0024313.jpg
ISIC_0024314.jpg
ISIC_0024315.jpg
```

Predefined 70/15/15 stratified splits were loaded. The training set exhibits class imbalance (~80% benign, ~20% malignant).

```
import pandas as pd

train_df = pd.read_csv("/content/drive/MyDrive/train.csv")
```

```
val_df   = pd.read_csv("/content/drive/MyDrive/val.csv")
test_df  = pd.read_csv("/content/drive/MyDrive/test.csv")

train_df.head()
```

	image_path	label
0	/content/HAM10000/archive (2)/HAM10000_images_...	0
1	/content/HAM10000/archive (2)/HAM10000_images_...	0
2	/content/HAM10000/archive (2)/HAM10000_images_...	1
3	/content/HAM10000/archive (2)/HAM10000_images_...	0
4	/content/HAM10000/archive (2)/HAM10000_images_...	0

Next steps: Generate code with train_df New interactive sheet

```
import os

for p in train_df["image_path"].sample(5):
    print(p, os.path.exists(p))
```

```
/content/HAM10000/archive (2)/HAM10000_images_part_1/ISIC_0027889.jpg True
/content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0034031.jpg True
/content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0033980.jpg True
/content/HAM10000/archive (2)/HAM10000_images_part_2/ISIC_0031523.jpg True
/content/HAM10000/archive (2)/HAM10000_images_part_1/ISIC_0028807.jpg True
```

```
print("Train size:", len(train_df))
print("Val size:", len(val_df))
print("Test size:", len(test_df))

print("\nLabel distribution (train):")
print(train_df["label"].value_counts(normalize=True))
```

```
Train size: 7010
Val size: 1502
Test size: 1503

Label distribution (train):
label
0    0.80485
1    0.19515
Name: proportion, dtype: float64
```

All images were resized to 224×224 pixels. Batch size was set to 32. TensorFlow AUTOTUNE was used for efficient data pipeline optimization.

```
IMG_SIZE = 224
BATCH_SIZE = 32
AUTOTUNE = tf.data.AUTOTUNE
```

Images were loaded from disk, decoded as RGB, resized to 224×224, and converted to float32 format before normalization within the preprocessing pipeline.

```
def preprocess_image(path, label):
    image = tf.io.read_file(path)
    image = tf.image.decode_jpeg(image, channels=3)
    image = tf.image.resize(image, (IMG_SIZE, IMG_SIZE))

    return image, label
```

On-the-fly data augmentation was applied to training images only, including horizontal flipping, brightness and contrast adjustments, 90° rotations, and mild zooming (±10%). This increases training diversity and reduces overfitting while preserving lesion morphology.

```
import numpy as np
import tensorflow as tf

def random_rotate(image):
    # Rotate by 0°, 90°, 180°, or 270°
    k = tf.random.uniform([], minval=0, maxval=4, dtype=tf.int32)
    return tf.image.rot90(image, k)


def random_zoom(image, zoom_range=(0.9, 1.1)):
    h, w = IMG_SIZE, IMG_SIZE
    zoom = tf.random.uniform([], zoom_range[0], zoom_range[1])
    new_h = tf.cast(h * zoom, tf.int32)
    new_w = tf.cast(w * zoom, tf.int32)
    image = tf.image.resize(image, (new_h, new_w))
    image = tf.image.resize_with_crop_or_pad(image, h, w)
    return image
```

```
def augment(image, label):
    image = tf.image.random_flip_left_right(image)
    image = tf.image.random_brightness(image, 0.1)
    image = tf.image.random_contrast(image, 0.8, 1.2)
```

```
    image = random_rotate(image)
    image = random_zoom(image)

    return image, label
```

After augmentation, images were normalized using EfficientNet's ImageNet-specific preprocessing function, which scales pixel values to the range [-1,1].

```
def normalize(image, label):
    image = tf.cast(image, tf.float32)
    image = tf.keras.applications.efficientnet.preprocess_input(image)
    return image, label
```

A TensorFlow tf.data pipeline was constructed to enable efficient streaming, augmentation, and normalization. Augmentation was applied only to training data, while validation and test sets underwent preprocessing and normalization only.

```
def df_to_dataset(df, training=False):
    ds = tf.data.Dataset.from_tensor_slices(
        (df["image_path"].values, df["label"].values)
    )

    if training:
        ds.shuffle(len(df), seed=42)

    ds = ds.map(preprocess_image, num_parallel_calls=AUTOTUNE)

    if training:
        ds = ds.map(augment, num_parallel_calls=AUTOTUNE)

    ds = ds.map(normalize, num_parallel_calls=AUTOTUNE)

    ds = ds.batch(BATCH_SIZE).prefetch(AUTOTUNE)
    return ds
```

To address class imbalance, cost-sensitive learning was implemented using inverse-frequency class weights computed from the training set.

```
from sklearn.utils.class_weight import compute_class_weight

weights = compute_class_weight(
    class_weight="balanced",
    classes=np.array([0,1]),
    y=train_df["label"].values
)

class_weights = {0: weights[0], 1: weights[1]}
print(class_weights)
```

```
{0: np.float64(0.6212336051045728), 1: np.float64(2.5621345029239766)}
```

Architecture:

EfficientNetB4 backbone

GlobalAveragePooling

Dense(256) + L2

Dropout(0.5)

Dense(1, sigmoid)

```
base_model = tf.keras.applications.EfficientNetB4(
    weights="imagenet",
    include_top=False,
    input_shape=(IMG_SIZE, IMG_SIZE, 3)
)

base_model.trainable = False  # STAGE 1

inputs = tf.keras.Input(shape=(IMG_SIZE, IMG_SIZE, 3))
x = base_model(inputs, training=False)
x = tf.keras.layers.GlobalAveragePooling2D()(x)
x = tf.keras.layers.Dense(
    256,
    activation="relu",
    kernel_regularizer=tf.keras.regularizers.l2(1e-4)
)(x)
x = tf.keras.layers.Dropout(0.5)(x)
outputs = tf.keras.layers.Dense(1, activation="sigmoid")(x)

model = tf.keras.Model(inputs, outputs)
model.summary()
```

```
Downloading data from https://storage.googleapis.com/keras-applications/efficien
71686520/71686520 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Model: "functional"
```

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0
efficientnetb4 (Functional)	(None, 7, 7, 1792)	17,673,823
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1792)	0
dense (Dense)	(None, 256)	459,008
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 1)	257

```
Total params: 18,133,088 (69.17 MB)
Trainable params: 459,265 (1.75 MB)
```

The model was compiled using the Adam optimizer with a learning rate of 1×10^{-4} and binary cross-entropy loss. Performance was monitored using AUC, precision, and recall.

```
model.compile(
    optimizer=tf.keras.optimizers.Adam(1e-4),
    loss="binary_crossentropy",
    metrics=[
        tf.keras.metrics.AUC(name="auc"),
        tf.keras.metrics.Precision(name="precision"),
        tf.keras.metrics.Recall(name="recall"),
    ]
)
```

Training employed early stopping and adaptive learning rate reduction based on validation AUC to prevent overfitting and ensure stable convergence.

```
callbacks = [
    tf.keras.callbacks.ModelCheckpoint(
        "best_model.keras",
        monitor="val_auc",
        mode="max",
        save_best_only=True,
        verbose=1
    ),
    tf.keras.callbacks.EarlyStopping(
```

```
            monitor="val_auc",
            mode="max",
            patience=5,
            restore_best_weights=True
        ),
        tf.keras.callbacks.ReduceLROnPlateau(
            monitor="val_auc",
            mode="max",
            factor=0.5,
            patience=2,
            verbose=1
        )
    ]
```

```
train_ds = df_to_dataset(train_df, training=True)
val_ds   = df_to_dataset(val_df, training=False)
test_ds  = df_to_dataset(test_df, training=False)
```

In Stage 1, the pretrained EfficientNet-B4 backbone was frozen and only the classification head was trained.

```
history_stage1 = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=30,
    class_weight=class_weights,
    callbacks=callbacks
)
```

```
━━━━━━━━━━━━━━━━━━━━ 48s 219ms/step - auc: 0.9030 - loss: 0.4311 - precision: 0.5013 - recall:

━━━━━━━━━━━━━━━━━━━━ 0s 191ms/step - auc: 0.9060 - loss: 0.4255 - precision: 0.4908 - recall: 0
ed from 0.88262 to 0.88295, saving model to best_model.keras
━━━━━━━━━━━━━━━━━━━━ 50s 229ms/step - auc: 0.9060 - loss: 0.4255 - precision: 0.4909 - recall:

━━━━━━━━━━━━━━━━━━━━ 0s 192ms/step - auc: 0.9106 - loss: 0.4172 - precision: 0.5028 - recall: 0
ed from 0.88295 to 0.88316, saving model to best_model.keras
━━━━━━━━━━━━━━━━━━━━ 85s 245ms/step - auc: 0.9105 - loss: 0.4173 - precision: 0.5028 - recall:

━━━━━━━━━━━━━━━━━━━━ 0s 191ms/step - auc: 0.9098 - loss: 0.4178 - precision: 0.5072 - recall: 0
ed from 0.88316 to 0.88401, saving model to best_model.keras
━━━━━━━━━━━━━━━━━━━━ 54s 245ms/step - auc: 0.9098 - loss: 0.4179 - precision: 0.5072 - recall:

━━━━━━━━━━━━━━━━━━━━ 0s 193ms/step - auc: 0.9132 - loss: 0.4122 - precision: 0.5058 - recall: 0
ed from 0.88401 to 0.88526, saving model to best_model.keras
━━━━━━━━━━━━━━━━━━━━ 51s 231ms/step - auc: 0.9132 - loss: 0.4122 - precision: 0.5058 - recall:

━━━━━━━━━━━━━━━━━━━━ 0s 192ms/step - auc: 0.9142 - loss: 0.4111 - precision: 0.5095 - recall: 0
t improve from 0.88526
━━━━━━━━━━━━━━━━━━━━ 53s 239ms/step - auc: 0.9142 - loss: 0.4111 - precision: 0.5096 - recall:

━━━━━━━━━━━━━━━━━━━━ 0s 197ms/step - auc: 0.9133 - loss: 0.4120 - precision: 0.5056 - recall: 0
t improve from 0.88526

eau reducing learning rate to 2.499999936844688e-05.
━━━━━━━━━━━━━━━━━━━━ 81s 235ms/step - auc: 0.9133 - loss: 0.4121 - precision: 0.5056 - recall:

━━━━━━━━━━━━━━━━━━━━ 0s 201ms/step - auc: 0.9200 - loss: 0.4038 - precision: 0.5119 - recall: 0
t improve from 0.88526
━━━━━━━━━━━━━━━━━━━━ 85s 248ms/step - auc: 0.9200 - loss: 0.4038 - precision: 0.5120 - recall:

━━━━━━━━━━━━━━━━━━━━ 0s 196ms/step - auc: 0.9166 - loss: 0.4059 - precision: 0.5153 - recall: 0
t improve from 0.88526

eau reducing learning rate to 1.249999968422344e-05.
━━━━━━━━━━━━━━━━━━━━ 54s 243ms/step - auc: 0.9166 - loss: 0.4060 - precision: 0.5153 - recall:

━━━━━━━━━━━━━━━━━━━━ 0s 193ms/step - auc: 0.9105 - loss: 0.4167 - precision: 0.5094 - recall: 0
t improve from 0.88526
━━━━━━━━━━━━━━━━━━━━ 78s 223ms/step - auc: 0.9105 - loss: 0.4166 - precision: 0.5095 - recall:
```

In Stage 2, the last 20 layers of the backbone were unfrozen and fine-tuned using a lower learning rate (1×10^{-5}).

```
base_model.trainable = True

for layer in base_model.layers[:-20]:
    layer.trainable = False
```

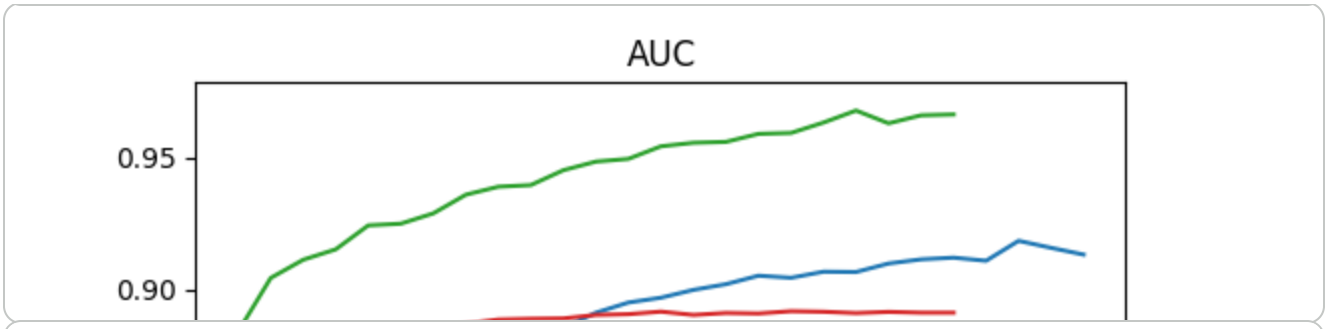
```
model.compile(
    optimizer=tf.keras.optimizers.Adam(1e-5),  # LOWER LR
    loss="binary_crossentropy",
    metrics=[
        tf.keras.metrics.AUC(name="auc"),
        tf.keras.metrics.Precision(name="precision"),
        tf.keras.metrics.Recall(name="recall"),
    ]
)
```

```
history_stage2 = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=30,
    class_weight=class_weights,
    callbacks=callbacks
)
```

```
Epoch 18: val_auc improved from 0.89172 to 0.89187, saving model to best_model
220/220 ━━━━━━━━━━━━━━━━━━━━ 84s 230ms/step - auc: 0.9593 - loss: 0.3047 - pre
Epoch 19/30
219/220 ━━━━━━━━━━━━━━━━━━━━ 0s 190ms/step - auc: 0.9628 - loss: 0.2950 - prec
Epoch 19: val_auc did not improve from 0.89187
220/220 ━━━━━━━━━━━━━━━━━━━━ 51s 228ms/step - auc: 0.9628 - loss: 0.2950 - pre
Epoch 20/30
219/220 ━━━━━━━━━━━━━━━━━━━━ 0s 193ms/step - auc: 0.9688 - loss: 0.2795 - prec
Epoch 20: val_auc did not improve from 0.89187

Epoch 20: ReduceLROnPlateau reducing learning rate to 2.499999936844688e-06.
220/220 ━━━━━━━━━━━━━━━━━━━━ 83s 231ms/step - auc: 0.9688 - loss: 0.2795 - pre
Epoch 21/30
219/220 ━━━━━━━━━━━━━━━━━━━━ 0s 194ms/step - auc: 0.9629 - loss: 0.2934 - prec
Epoch 21: val_auc did not improve from 0.89187
220/220 ━━━━━━━━━━━━━━━━━━━━ 49s 224ms/step - auc: 0.9629 - loss: 0.2934 - pre
Epoch 22/30
219/220 ━━━━━━━━━━━━━━━━━━━━ 0s 192ms/step - auc: 0.9664 - loss: 0.2850 - prec
Epoch 22: val_auc did not improve from 0.89187

Epoch 22: ReduceLROnPlateau reducing learning rate to 1.249999968422344e-06.
220/220 ━━━━━━━━━━━━━━━━━━━━ 51s 230ms/step - auc: 0.9664 - loss: 0.2850 - pre
Epoch 23/30
219/220 ━━━━━━━━━━━━━━━━━━━━ 0s 191ms/step - auc: 0.9686 - loss: 0.2777 - prec
Epoch 23: val_auc did not improve from 0.89187
220/220 ━━━━━━━━━━━━━━━━━━━━ 82s 229ms/step - auc: 0.9686 - loss: 0.2777 - pre
```

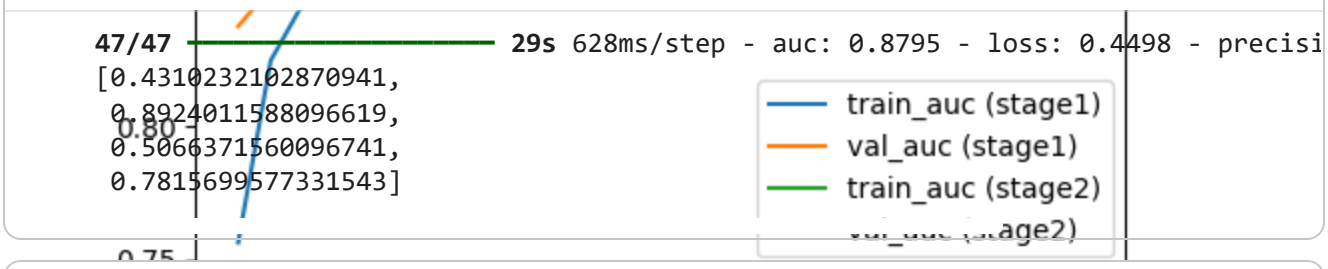
Training and validation curves were plotted to analyze convergence behavior and overfitting dynamics across both stages.

```
import matplotlib.pyplot as plt
def plot_metric(h1, h2, metric):
    plt.figure(figsize=(6,4))
    plt.plot(h1.history[metric], label=f"train_{metric} (stage1)")
    plt.plot(h1.history[f"val_{metric}"], label=f"val_{metric} (stage1)")
    plt.plot(h2.history[metric], label=f"train_{metric} (stage2)")
    plt.plot(h2.history[f"val_{metric}"], label=f"val_{metric} (stage2)")
    plt.legend()
    plt.title(metric.upper())
    plt.show()

plot_metric(history_stage1, history_stage2, "auc")
plot_metric(history_stage1, history_stage2, "recall")
plot_metric(history_stage1, history_stage2, "precision")
```



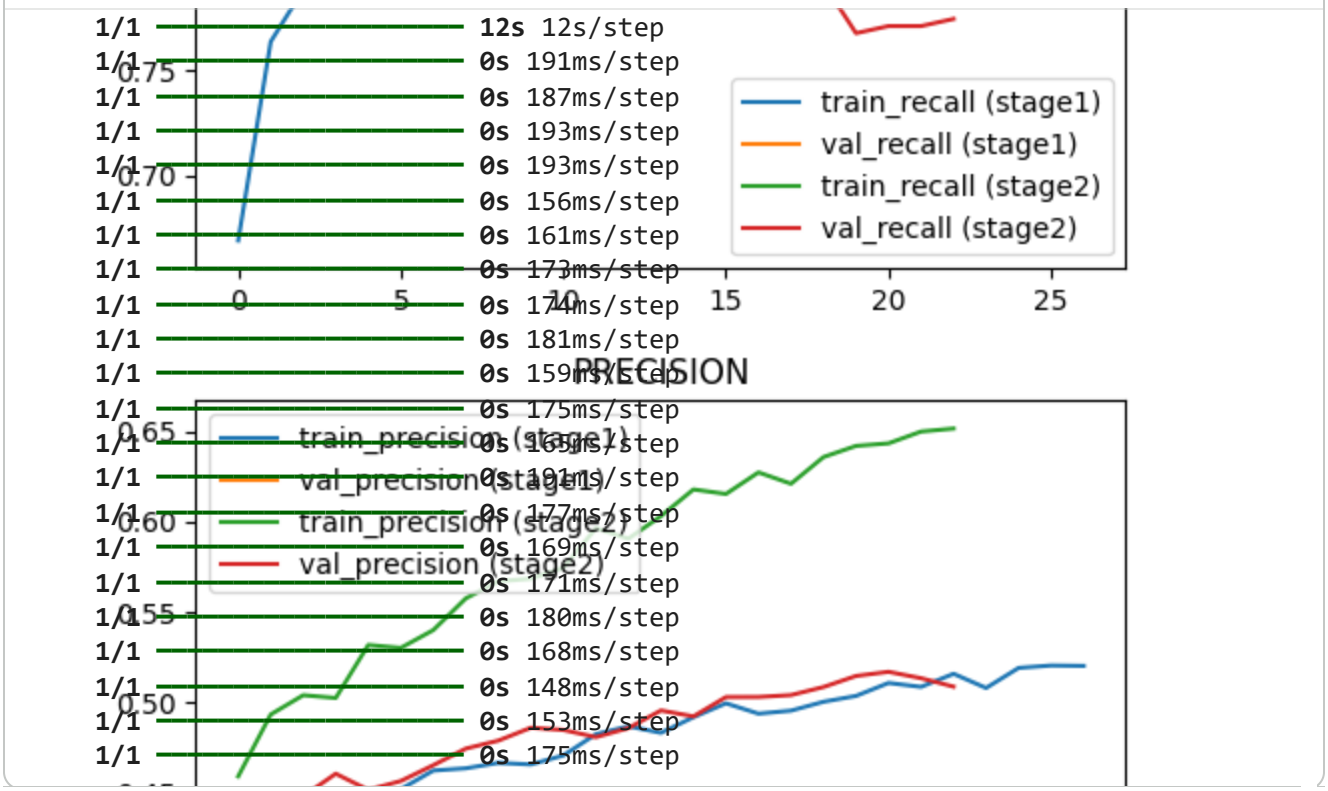

```
model.evaluate(test_ds)
```

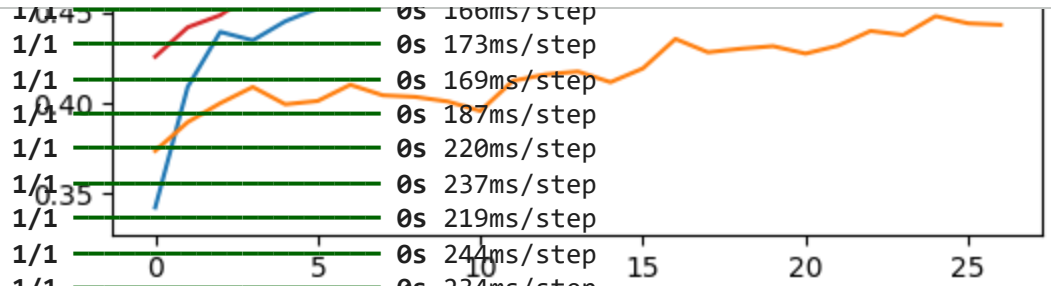
```
47/47 ━━━━━━━━━━━━━━━━━━━━ 29s 628ms/step - auc: 0.8795 - loss: 0.4498 - precisi
[0.4310232102870941,
 0.8924011588096619,
 0.5066371560096741,
 0.7815699577331543]
```




```
from sklearn.metrics import classification_report, confusion_matrix

y_true, y_pred = [], []

for x, y in test_ds:
    preds = model.predict(x)
    y_true.extend(y.numpy())
    y_pred.extend((preds > 0.5).astype(int))

print(classification_report(y_true, y_pred, target_names=["Benign", "Malignant"
print(confusion_matrix(y_true, y_pred))
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 12s 12s/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 191ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 187ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 193ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 193ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 156ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 161ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 173ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 174ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 181ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 159ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 175ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 165ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 191ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 177ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 169ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 171ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 180ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 168ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 148ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 153ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 175ms/step
```






```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 166ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 173ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 169ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 187ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 220ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 237ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 219ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 244ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 234ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 296ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 261ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 270ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 264ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 233ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 269ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 234ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 185ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 160ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 179ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 165ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 164ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 173ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 126ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 118ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 12s 12s/step
              precision    recall  f1-score   support

      Benign       0.94      0.82      0.87      1210
   Malignant       0.51      0.78      0.61       293

    accuracy                           0.81      1503
   macro avg       0.72      0.80      0.74      1503
weighted avg       0.85      0.81      0.82      1503

[[987 223]
 [ 64 229]]
```

```
import numpy as np
from sklearn.metrics import roc_curve, confusion_matrix

# Get y_true and y_prob from val_ds
y_true_val = np.concatenate([y.numpy() for _, y in val_ds]).astype(int).ravel()
y_prob_val = model.predict(val_ds).ravel()

fpr, tpr, thr = roc_curve(y_true_val, y_prob_val)

target_recall = 0.90
idx = np.where(tpr >= target_recall)[0][0]
best_thr = thr[idx]

print("Chosen threshold (val):", best_thr)
print("Val recall at threshold:", tpr[idx])
```

```
47/47 ━━━━━━━━━━━━━━━━━━━━ 33s 458ms/step
Chosen threshold (val): 0.30874923
Val recall at threshold: 0.9010238907849829
```

```
y_true_test = np.concatenate([y.numpy() for _, y in test_ds]).astype(int).ravel
y_prob_test = model.predict(test_ds).ravel()

y_pred_test = (y_prob_test >= best_thr).astype(int)

tn, fp, fn, tp = confusion_matrix(y_true_test, y_pred_test).ravel()

accuracy = (tp + tn) / (tp + tn + fp + fn)
precision = tp / (tp + fp) if (tp + fp) else 0.0
recall = tp / (tp + fn) if (tp + fn) else 0.0
specificity = tn / (tn + fp) if (tn + fp) else 0.0
f1 = 2*precision*recall/(precision+recall) if (precision+recall) else 0.0

print("Test confusion matrix:", [[tn, fp],[fn, tp]])
print(f"Accuracy: {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall: {recall:.4f}")
print(f"Specificity: {specificity:.4f}")
print(f"F1: {f1:.4f}")
```

```
47/47 ━━━━━━━━━━━━━━━━━━━━ 16s 330ms/step
Test confusion matrix: [[np.int64(881), np.int64(329)], [np.int64(34), np.int64(
Accuracy: 0.7585
Precision: 0.4405
Recall: 0.8840
Specificity: 0.7281
F1: 0.5880
```

```
import pandas as pd

results = {
    "Model": "EfficientNet-B4",
    "AUC": np.max(tpr),
    "Recall": recall,
    "Precision": precision,
    "Specificity": specificity,
    "F1": f1,
    "Threshold": best_thr
}

pd.DataFrame([results]).to_csv("final_metrics_efficientnetB4.csv", index=False)
```

```
import matplotlib.pyplot as plt
import numpy as np

# -------- SETTINGS --------
threshold = best_thr
num_images = 9

# Collect test images and labels
images_list = []
labels_list = []

for images, labels in test_ds.take(1):
    images_list = images.numpy()
    labels_list = labels.numpy()
    break

# Predict probabilities
probs = model.predict(images_list).ravel()

plt.figure(figsize=(15, 12))

for i in range(num_images):
    prob = probs[i]
    true_label = "Malignant" if labels_list[i] == 1 else "Benign"
    pred_label = "Malignant" if prob >= threshold else "Benign"

    # Reverse EfficientNet preprocessing for display
    img = images_list[i]
    img_display = (img - img.min()) / (img.max() - img.min())

    plt.subplot(3, 3, i + 1)
    plt.imshow(img_display)
    plt.axis("off")

    plt.title(
        f"True: {true_label}\n"
        f"Pred: {pred_label}\n"
        f"P(malignant): {prob:.2f}",
        fontsize=10
    )

plt.suptitle("EfficientNet-B4 Test Predictions (9 Samples)", fontsize=16)
plt.tight_layout()
plt.show()
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 186ms/step
```

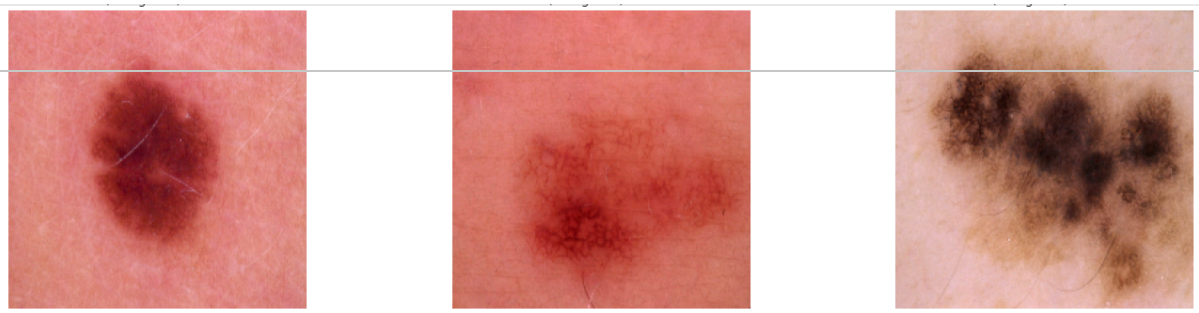
```
model = tf.keras.models.load_model("best_model.keras")
```

```
model.evaluate(test_ds)
```

```
47/47 ━━━━━━━━━━━━━━━━━━━━ 28s 309ms/step - auc: 0.8795 - loss: 0.4498 - precisi
[0.4310232102870941,
 0.8924031158096619,
 0.5066371560096741,
 0.7815699577331543]
```

```
model.save("final_efficientnetB4_skin_cancer.keras")
```

```
from google.colab import files
files.download("final_efficientnetB4_skin_cancer.keras")
```



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